

Acme Corp - Nexus Integration Hub

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Deployment Guide

Prerequisites

- Docker 24+ and Docker Compose v2
- Python 3.11+
- Access to OAuth provider credentials (see authentication_guide.pdf)
- AWS credentials for S3 artifact storage

Local Development

```
git clone https://github.com/acme-corp/nexus-integration-hub.git
cd nexus-integration-hub
cp .env.example .env
docker compose up -d postgres redis kafka
pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000
```

Environment Variables

DATABASE_URL — PostgreSQL connection string (required)
REDIS_URL — Redis connection string (required)
OAUTH_CLIENT_ID — OAuth client ID (required)
OAUTH_CLIENT_SECRET — OAuth client secret (required)
S3_BUCKET — S3 bucket for sync artifacts (required)
KAFKA_BOOTSTRAP_SERVERS — Kafka brokers (required)

Staging Deployment

Staging deploys automatically on merge to main via GitHub Actions.

Push to main, CI runs tests and builds Docker image, ArgoCD syncs to staging cluster.

Production Deployment

Production requires manual approval in GitHub Actions with two approvers.

Blue-green deployment with automatic rollback on health check failure.

Rollback Procedure

```
kubectl rollout undo deployment/nexus-integration-hub -n production
```

Or use admin API POST <https://api.nexus.acme-corp.example.com/admin/rollback>

Monitoring Post-Deploy

Verify /health returns 200, /ready returns 200, Prometheus metrics flowing,
no error rate spike in Grafana, OAuth token flow works end-to-end.

Known Issues

Helm chart 1.4.x has Redis password injection bug — use 1.5.0+

Celery worker memory leak fixed in v2.4.0; do not deploy older versions